Appendix 1

Efficient and inefficient markets

Contents

A1.1. Market efficiency 2
A1.1.1. Efficient market hypothesis 2
A1.1.2. Evidence from the markets disproving the EMH 3
A1.1.3. Key paradigms in traditional finance 4
A1.2. A brief account of behavioral finance 5
A1.2.1. How people make decisions 5
A1.2.2. Heuristics and behavioral finance 6
A1.2.3. Themes in behavioral finance 6
A1.2.4. Behavioral finance and investment behavior 11
A1.2.5. Imperfections in the markets 13
A1.2.6. Response from traditional finance 14
A1.2.7. Lessons learned 16
A1.3. Recent models with behavioral finance considerations 16
A1.3.1. Goal-based wealth management 16
A1.3.2. Behavioral portfolio theory (BPT) 18
A1.3.3. Popularity asset pricing model 18
Additional references 19
Appendix: Questionnaire on behavioral finance 21

Author: Ka Lok Chau

Last Revised: January 2024

A1.1. Market efficiency

A1.1.1. Efficient market hypothesis

Classical finance models, such as MPT and CAPM, assume that stock prices are random. An important formulation of such phenomenon called the *Efficient Market Hypothesis* (EMH) was proposed by Professor Eugene Fama of the University of Chicago in the 1960's (who won the Nobel Prize in Economics in 2013). Stock prices follow a random walk, i.e. price changes are random and unpredictable. In other words, current stock prices in the market reflect all available information, and stock prices would only change because of the arrival of new information. Two colorful examples from an influential book "A Random Walk down Walk Street" by Princeton professor Burton Malkiel can illustrate this concept:

- A blindfolded chimpanzee throwing darts at the Wall Street Journal could select a portfolio that would do as well as the experts;
- A finance professor and a student walked along a street and the student found a US$100 bill lying on the ground. As the student stops to pick it up, the professor says, "Don't bother – if it were really a $100 bill, it wouldn't be there."

When no new information is available and the market is in equilibrium, the only way to earn a higher return is to take more risk. If the market is efficient, essentially it means that *no investor could earn above-average risk-adjusted returns.*

Implications of the EMH

There are three versions of the EMH, with increasingly strong assumptions:

- **Weak form**: Stock prices already reflect all information that could be derived from past trading information, e.g. historical prices and trading volume;
- **Semi-strong form**: Stock prices reflect all publicly available information, including price history, company's result forecasts, etc.;
- **Strong form**: Stock prices represent all publicly and privately available information (note that insider trading is not allowed in many countries).

Market professionals usually rely on two approaches when they trade stocks:

- **Fundamental analysis**: Analysts would examine a company's past records, financial situation, current economic environment, and other external factors to determine whether the company's stock is worth investing in.
- **Technical analysis**: Analysts (known as "chartists") believe that historical price series could generate buy/sell signals. A popular theory is the Elliott Wave Theory, which claims that stock prices can be described by a set of wave patterns.

If semi-strong form of EMH is correct, fundamental analysis would be useless because it would not be possible to obtain any advantage by studying current information as these information are available to anyone. Equity research analysts may argue that their research would be of benefit because the analysis is *better* than other people's analysis. Even if only the weak form of EMH is correct, technical analysis should be useless as everyone has the same tool and would have taken action until the stock prices reach an equilibrium. As a result, it would be logical to invest in a market portfolio, e.g. an index

tracking fund, instead of paying for the service of fund managers and analysts. It is not surprising that *EMH is unpopular among professionals!*[1]

Testing the EMH

The claims of the EMH have been controversial from the start and numerous tests were developed to verify the validity of the theory. A central thesis is that no one can consistently beat the market on a risk-adjusted basis (also see section below). If a successful trading strategy can be identified and shown to be a consistent winning strategy, then it would be a strong rebuttal of EMH. However, meaningful test results are difficult to obtain because profit/loss are usually small with respect to the trading size, and thus the results may not be statistically significant. Furthermore, the really successful strategies are seldom published (a selection bias issue). Finally, some trading profits are due to luck and may not be repeatable in the future.

Weak-form tests usually start with calculating the autocorrelation among security returns using historical price series. Small autocorrelation is often noticeable, but insignificant. Some kinds of trading filters are used to generate buy/sell signals; again small but insignificant gains could be achieved. However, these tests are not conclusive. Fama, Fisher, Jensen and Roll (1969) proposed tests for the semi-strong form of EMH. Various market information such as stock split, share repurchase, and recommendations by analysts dividend announcements have been collected so that trading strategies can be formulated based on these information, but it was found that little extra return could be generated. Finally, some tests were carried out to examine the investment performance of individuals or groups who are in a position to possess nonpublic information, as well as the performance of major market participants. Results show that the market is probably not strong form efficient.

A1.1.2. Evidence from the markets disproving the EMH

Performance of fund managers

During the early days of the debate of the validity of the EMH, the example of Peter Lynch, manager of Fidelity Magellan Fund from 1977-1990, was often mentioned. During this period, his fund had consistently beat the market every year (except for 2 years), and often by a wide margin. In numeric terms, $1000 invested in 1977 became $28,000 in 1990, representing an annual return of 29.2% p.a., whereas S&P 500 has returned less than 14% p.a. in the equivalent period. It sparked a wide debate as to whether the market is efficient and whether some people are more skillful than others.

Percentage of Active Funds Outperformed by Benchmarks (Malkiel (2019), p.177)			
	1 year	5 Years	15 years
All Large Cap funds vs S&P500	63.08	84.23	92.33
All Small Cap funds vs S&P Small cap 600	47.70	91.17	95.73
Global Funds vs S&P Global 1200	50.21	77.71	82.47
Emerging Market Funds vs S&P IFCI Composite	64.89	77.78	94.83

[1] Professor Fama has co-established a fund management company in 1981 and remained as a board Member and consultant.

Those who support the EMH will quote figures such as the table above, showing that most fund managers cannot outperform broad-based market indices. Some managers have good performance in one year but perform badly in subsequent years, probably indicating that their performance could be due to luck ("Even a broken clock could be right twice a day").[2]

Survivorship bias
In testing mutual fund returns, we have to remember that many mutual fund "died" and long term records were not kept. A particularly prevalent marketing strategy is to merge a weak performing fund into other funds, so that only funds with comparatively better track records would be available for investment. In other words, the surviving funds that have long term records are those which perform reasonably well. Hence any conclusion about funds performance could be biased upwards

Market anomalies
Research has shown that there are apparent evidences of market events which contradict with the EMH. (A word of warning in interpreting the results: we need to use risk-adjusted returns in testing the EMH, but it is not a trivial task to perform the correct risk adjustment procedure.)

- *P/E effect*: Stocks with low Price/Earning ratio (P/E ratio) outperform those with high P/E.
- *Firm size effect*: Small companies outperform bigger companies.
- *Day-of-week effect*: Returns are the lowest on Mondays.
- *Year-end effect*: Investing January produces the highest return as stock prices are depressed at year end (often due to tax considerations).
- *Dividend effect*: stocks with high dividend yields produce higher returns.
- *Momentum effect*: e.g. a positive return in one week is more likely to be followed by a positive return in the next week, i.e. prices show positive serial correlation.
- *Excess volatility*: market movements could not be explained by changes in predictions of earnings or dividends.

A1.1.3. Key paradigms in traditional finance

According to Lo (2017), traditional investment paradigm can be summarized as five principles, which are the accepted wisdom for investment professionals and academia.

Principle 1: The Risk/Reward trade-off. Assets that have higher returns must bear additional risk.
Principle 2: Equilibrium return is governed by the Capital Asset Pricing Model - i.e. risk of an asset is related by its beta.
Principle 3: Portfolio optimization is performed utilizing the Modern Portfolio Theory and by passive investment through index funds.
Principle 4: The importance of asset allocation, i.e. deciding how much to invest in each asset class, compared to picking securities from each asset class.
Principle 5: Investors should hold mostly equities for the long run.

[2] Further arguments can be found in Professor Malkiel's book, *A Random Walk down Wall Street.*

Since the start of the 20th century, these principles worked very well and investors generally profited from related strategies. However, beginning from the 21st century, many people began to question whether these theories still work or not, and many explanations have been provided.

A1.2. A brief account of behavioral finance

A1.2.1. How people make decisions

We have introduced a few different functional forms for utility functions in section 5.3.5. In particular, we mention that utility functions should have a concave shape as utility should not decrease when wealth increases. However, as early as 1948, Friedman and Savage noticed that many people exhibit "strange" behavior, where they buy insurance and lottery tickets at the same time. Buying insurance is equivalent to accepting a *certain* loss of a small sum (the insurance premium) in preference to a combination of a small chance of a much larger loss (when disaster struck) and a large chance of no loss; i.e. prefer certainty to uncertainty. On the other hand, when one buys a lottery ticket, he/she is accepting a large chance of a small loss (the price of the ticket) plus a small chance of winning a large amount, in preference to avoiding both risks (by refusing to buy the ticket). Essentially the person is choosing uncertainty in preference to certainty. The standard concave utility function defined over final wealth cannot explain this behavior.

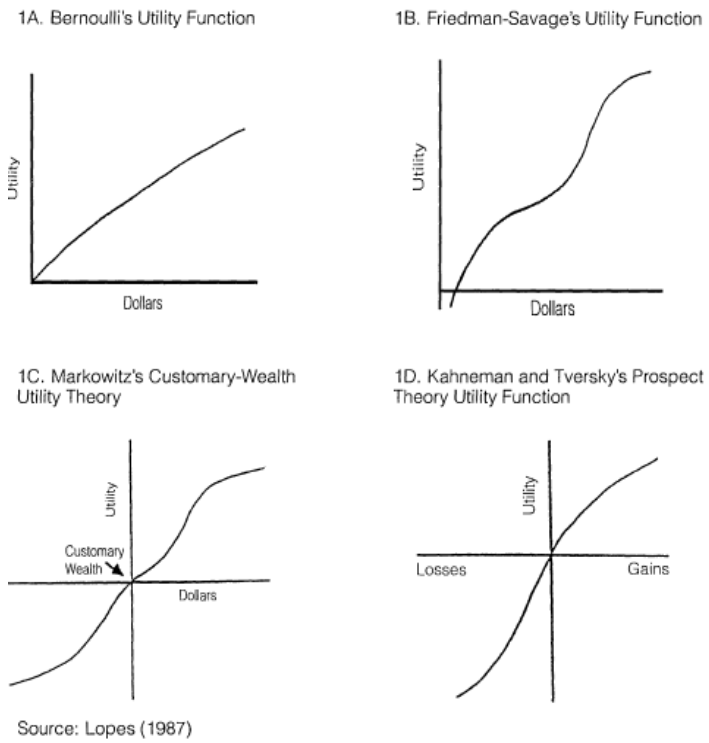


Figure taken from Shefrin and Statman (2000), quoting from Lopes (1987).

In order to resolve this puzzle, many authors suggest more complex types of utility functions, such as those shown in the figure above. (We would look at Kahneman and Tversky's function in more detail in the next section). A key development took place in the 1980s. Whereas traditional finance seeks to understand financial markets using models in

which agents are "rational," a new area known as behavioral finance emerged, in response to the difficulties faced by the traditional paradigm. In particular, these theories try to take into account that some agents are not fully rational, and aim to specify the form of irrationality in order to incorporate these insights into a consistent framework.

A1.2.2. Heuristics and behavioral finance

Behavioral finance can be seen as *an application of psychology to financial behavior*. Following Shefrin (2000), there are three themes in behavioral finance:

- *Heuristic-driven bias*: Finance practitioners commit errors based on "rules of thumb."
- *Framing problem*: people make decisions based on how the problem is presented.
- *Inefficient markets*: the wrong decisions made by people contribute to inefficiency in markets, which means prices are different from their fundamental values.

We will look at examples of the last two themes below. Heuristics is an important concept in behavioral finance. Many studies show that people are incapable of handling complex situations as one may not be able to assimilate all the available information simultaneously. People tend not to analyze each situation rationally, but they develop general principles for themselves to be applied to many different scenarios. However, these rules are usually imperfect, and people commit errors in particular situations.[3]

A1.2.3. Themes in behavioral finance

Many themes in behavioral finance are best illustrated with simple examples. We will look at their application in real markets in the next section.

The primacy effect

Assume that a low unemployment rate and a strong growth of non-farm payroll jobs are both desirable. If we compare the following sentences:

- *The US unemployment rate is at a historical low but growth of non-farm payroll jobs is less than expected.*
- *The growth of non-farm payroll jobs is less than expected but US unemployment rate is at a historical low.*

Both sentences give essentially the same information, but the majority of market participants are likely to judge the economic situation in the USA considerably more positively from the second statement. It shows that it is possible to influence people's perception by way of how information is ordered.

The endowment effect

People often demand much more to sell an object that they own than they would be willing to pay to buy it. For example, you have bought a Mark Six lottery ticket and have memorized the numbers that you bought. Would you be willing to sell the ticket at the original price to someone else? Rationally speaking, the cost of the ticket should not be affected by who is selling it.

[3] For a lively discussion of many of these themes, refer to the popular level book *Thinking, Fast and Slow* (Allen Lane, 2011) by Professor Daniel Kahneman.

Gambler's fallacy
At the roulette table in a casino, after a series of 9 red numbers in a row, what is the probability that the next number is red? ["The next number MUST be black"]. If the game is fair, of course the correct answer is 0.5. However, people do not realize the correct probability of this event and that the seemingly abnormal result could be due to probability (1 in 2^9, or once in every 512 games).

Representativeness
People make judgments based on familiar types, while disregarding the base rate probability. Also, they tend to see patterns in events which are necessarily random.[4] Two probability quizzes can illustrate this problem. The first example comes from a mathematical concept called the Law of Small Numbers.

- It is given that the mean IQ of Form 1 students is 100. You have selected a random sample of 50 children and the first child tested has an IQ of 150. What do you expect the mean IQ to be for the whole group of 50 children?

A surprising high number of people answer 100, because "in a random sample, high IQ would be neutralized by someone with low IQ." The correct analysis should be: one child has an IQ of 150, 49 children would have an average IQ of 100, hence the mean IQ of the group of 50 students is (150+49x100)/50 = 101. A second example shows the insensitivity to prior probability.

- You are told that there are 70 engineers and 30 lawyers in a group. You are then told that a person from this group "wear a suit to go to work, like to read English novels." What is the probability that this person is a lawyer?
- What if you are told originally that there are 30 engineers and 70 lawyers in the group? Would your answer be different?

Most people give more or less the same answer when they are asked both questions, whereas the correct answer should take into account of the distribution of the number of engineers and lawyers in the group.

Anchoring
People are divided into two groups, and each group has been assigned a random number from 1 to 100. In our example, group 1 gets a number 10, and group 2 gets a number 65. People in the two groups were asked whether the % of African countries in the United Nations is higher or lower than the given number, and what would be their estimate. A typical test result is as follows: in group 1, the average estimate is 25%; in group 2, the average estimate is 45%. Most people have no idea of the correct answer, and they use a convenient number as a reference (such as the random number given in the question). [The correct answer in this question is around 25%]. Hence it may be possible to influence a person's choice by supplying some kind of "nudge."

Illusion of control
Many people have a false sense of control when the reality is the opposite. Consider the following two situations:

- Situation A: a coin will be tossed and you are asked to bet whether it is head or tail.

[4] Refer to a series of books by the author Nassim Taleb, such as *Fooled by Randomness* (Random House, 2004).

- Situation B: a coin has already been tossed but you don't know the outcome; you are asked to bet whether it is head or tail.

According to many surveys, more people will bet in situation A than situation B.

Allais Paradox

Which do you prefer:

- A) A sure gain of $3000; or
- B) 80% chance of winning $4000?

Which do you prefer:

- C) 25% chance of winning $3000; or
- D) 20% chance of winning $4000?

(A) and (B) are just 4 times the probabilities of (C) and (D). If one prefers (A) to (B), the same person should prefer (C) to (D). In reality, most people choose (A) and (D), and these choices violate the substitution axiom of expected utility theory. One explanation of this phenomenon is that people put more emphasis on "certainty" of outcome, thus having a sure gain is preferable to a higher expected winning (although the probability of winning is still high). When the chance of winning is small, expected winning matters more.

Cognitive Dissonance

When someone learns that he/she has made a wrong decision, a mental conflict results. To reduce the possibility of this conflict (known as cognitive dissonance), people will avoid new information or develop unreasonable arguments to maintain their beliefs. For example, after the purchase is completed, new car purchasers will selectively avoid reading advertisements for car models that they did not choose, because they fear that they may have made a bad decision as they have missed some better deals (would you do this when you buy a new phone?)

Disposition Effect

- You have two stocks and each is worth $10000 now. However, stock A was bought at $8000 and stock B was bought at $11000. You want $10000 cash now. Which stock would you sell?

Most people sell stock A, because it validates a "good" decision to purchase A. People enjoy the pride in realizing a profit. Selling B means realizing a bad decision and people will feel the pain of regret.

Regret Avoidance

- You have bought the same six numbers for the Mark Six lottery for the past 2 years without winning any prize. How would you feel in the following two situations:
    - You keep the same numbers this week, but again you lose. Regret may be felt and this is called the *regret of omission.*
    - You switch to a new set of numbers but the old set won. This is called the *regret of commission.*

A regret of commission is more painful than a regret of omission. This can be one of the reasons why people are reluctant to make changes.

Framing of acts

People could not process more than one piece of information at a time. Let's consider the following example. Which do you prefer:

- A) a sure gain of $2400.
- B) 25% chance to gain $10000, 75% chance to gain 0.

In another game, you have to choose between:

- C) a sure loss of $7500.
- D) 75% chance to lose $10000, 25% chance to lose 0.

In a third game, the choices are:

- E) 25% chance to win $2400, 75% chance to lose $7600.
- F) 25% chance to win $2500, 75% chance to lose $7500.

When these games are presented individually, most people choose (A), (D) and (F). The strange thing is, (E) = (A)+(D), and (F) = (B)+(C) !!! If someone chooses (A) and (D) over (B) and (C), it is logically to assume that (E) will be chosen, but the result shows otherwise. It indicates that the outcome may be affected by how the acts are presented.

Another example concerns the framing of outcomes. Assume a new disease has broken out which has infected 600 people in a village. In one version, you have to choose between these options:

- Vaccine A – a certainty of saving 200 lives.
- Vaccine B – 67% chance that everyone will die, 33% chance that everyone is saved.

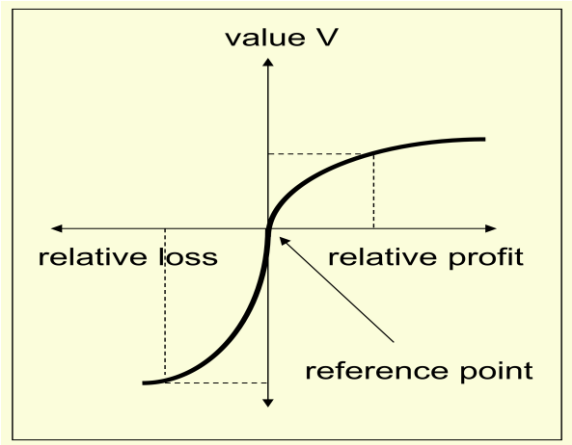
The choices can also be presented in an alternative manner.

- Vaccine C – 400 out of the 600 people will die.
- Vaccine D – 33% chance that nobody will die, 67% chance that everyone will die.

With these wordings, most people choose A and D. Obviously A and C are equivalent, but people prefer to see a certain number of people "saved" instead of a significant percentage will "die."

Prospect theory

Building on their joint work over many years, professors Daniel Kahneman and Amos Tversky, both psychologists, published a famous paper in the economics journal *Econometrica* in 1979, which has become the most cited paper of all time. This paper on Prospect Theory contains a number of insights. They claim that the decision maker is not concerned with final total wealth, but with changes in wealth relative to some reference point. People suffer more from a loss compare to the pleasure of a gain (with respect to the reference point). Hence the shape of the utility function would look different when comparing gains and losses.



In other words, the sensitivity of the value function changes across the reference point. Value increases slower when a relative profit is made, compared to a relative loss (the "reflection effect").[5]

The point of reference plays a major role in deciding the sensitivity, but sometimes it is difficult to decide what the reference point is. One obvious choice is to use the current wealth as the benchmark, but other points of reference are possible. Consider the following example:

- You bought some shares of Tencent at $250 three years ago. Last year, the price of the stock had risen to $700. You have seriously considered selling the stock but did not do so. Now the price is at $380, and you decide to sell. Would you consider that you have made a profit of $130, or suffer a loss of $320?

Economically speaking, a profit has been made, but many people look at the historical 52-week high as the reference (of course it is impossible to know when the high is reached without hindsight).

Loss aversion

The perception of gains and losses can also affect investment choices. Consider the following options:

- A) A sure gain of $3000.
- B) 80% chance of winning $4000, 20% chance of winning $0.

In a different game, the choices are:

- C) 80% chance of losing $4000, 20% chance of losing $0.
- D) A sure loss of $3000.

According to some surveys, most people choose (A) and (C). Apparently people are often risk averse when facing gains, but become risk takers when facing losses. This provides support to the S-shaped utility function as predicated by the prospect theory (although it is still difficult to determine the reference point).

Mental Accounting

Consider the following situations:

- Imagine that you decided to see a play where a ticket costs $500. As you enter the theatre you discovered you have lost a $500 note. Would you still pay $500 for a ticket to see the play?

[5] This S-shaped utility function is the representative "logo" of Prospect Theory.

- Imagine that you bought the $500 ticket yesterday. Just when you enter the theatre today, you discover that you have lost the ticket. Would you still pay $500 for another ticket?
- Would your answer be different if you bought the ticket 6 months ago? (*sunk cost effect*)

People often place different events in separate "mental accounts." Whereas the financial outcomes in both scenarios are equivalent – a loss of $500, or a cost of $1000 for the ticket, a typical answer is that in the first scenario, the person would buy another ticket because the ticket is still seen as costing $500. In the second scenario, it appears that the ticket "costs" $1000, which may seem to be too expensive.

Daniel Kahneman, the Nobel Prize winner who developed many of these theories, also suffered from mental accounting (from Bernstein (1996)). He and Amos Tversky found that a loss is less painful when it is just an addition to a larger loss, e.g. if you have already lost $1000 in your stock A, losing another $500 is less painful than if you suffer the $500 lost in another stock B. Thus when Kahneman moved to a new home, he and his wife bought all the furniture within a week after buying the house, where the furniture costs and the house cost are grouped together into a "house account." If they had looked at the furniture as a separate account, they might think it is too expensive and buy fewer pieces!

A1.2.4. Behavioral finance and investment behavior

We could observe elements of behavioral issues in all areas of finance, and some examples are given in the sections below.

Over-confidence and trading

People are often over-confident with their skills. In a survey in Sweden, it was found that over 80% of people claimed that they were "better than average" drivers (by definition, only 50% should be better than average). The same attitude can be seen in the stock market. Even without extra information, people believe the stocks they own will perform better than stocks they do not own (an illusion of control). Furthermore, investors tend to over-estimate their ability to interpret information and trade too often (see a famous study by Barber and Odean (2000)). On the other hand, many people would sell low and buy high – they hold on to losers for too long, while selling winners too quickly. Using the themes of behavioral finance, this kind of behavior can be explained by:

- *disposition effect* – people are unwilling to realize loss.
- *cognitive dissonance* – people want to make winning decisions.
- *endowment effect* – people are reluctant to sell something that they already own at a bad price.

Mutual fund selection

A typical report in almost all literature relating to mutual fund performance will show its results for the last 1, 3 and 5 years. This is what most people would look at in assessing the performance of the fund. However, even if a single fund has performed best in all these categories and the manager's skill could not be ignored, it is still possible that the result could be due to "luck." Consider the following example. Every month, a child will toss a coin and if the result is a head, he/she will predict the market will go up. If 4096 ($=2^{12}$) children perform this exercise and each child produces a different forecast, one child will end up being right every time for 12 consecutive months. Is he/she a "star" analyst?

The dividend puzzle

People like to receive dividends from stock investments, even though sometimes it is to their disadvantage (due to tax reasons). Theoretically, one could sell some shares to raise the same amount of cash as if they have received a dividend, but people don't do that. We could explain this phenomenon by the concepts of mental accounting and regret avoidance:

- People lack self control, and would only be willing to spend "income" rather than "capital."
- Income would not be confused with assets (the "piggy bank").
- If you sell some stock to raise cash, and the stock subsequently rises by 20% in a week, one would feel deep regret. In order to avoid regret, it is better not to take any action.

Fund manager buying familiar stocks

A fund manager has the following two choices:

- buy shares in Tencent (let's say the current valuation is attractive); or
- buy shares in stock X, a start-up company – if all goes well, this company should have earned a lot of money.

Subsequently if share price comes down and the portfolio loses money, which outcome will generate less regret (especially when faced with dis-satisfied customers)? If the manager buys Tencent and it has come down, one could blame the "market conditions." But if one buys stock X and it has come down, there is no one to blame but his own bad decision.
The answer is obvious – in order to avoid regret, big portfolios often need to buy the popular stocks.[6]

Forecasts from stock analysts

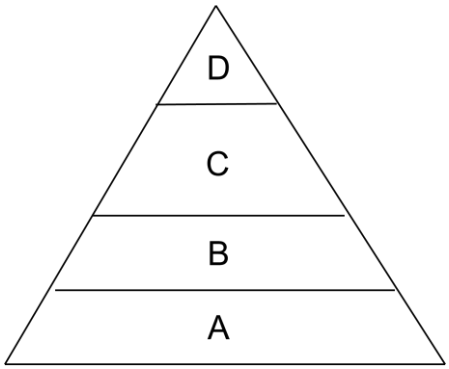
Stock prices often react to new research reports produced by stock analysts, especially from those who work in the major investment banks. However, de Bondt and Thaler (1985) demonstrated that people often over-react on new information and underweight prior and longer term information. As a result, stock prices systematically overshoot very far in either direction. Even stock analysts tend to ignore the long term trends, as it is often seen that big changes are made in analysts' forecasts after earning announcements from the relevant companies.

Stock analysts are human beings and many would also suffer from behavioral traits, such as:

- gambler's fallacy (e.g. stock market has gone down 3 years in a row; it would definitely go up this year).
- over-confidence, therefore they often get surprised.
- anchoring (e.g. analysts often naively project trends that they perceive), and the predictions are often skewed (i.e. their best guesses do not lie mid-way between their low and high guesses).

[6] Many years ago, Peter Lynch, the famous fund manager of the Magellan Fund, said that his fund would always include shares of IBM exactly because of this reason.

Portfolio construction



A typical asset pyramid, where each portion represents a different portfolio.

- A: to preserve wealth – safe investments like savings account.
- B: income generation – bonds, high dividend stocks etc.
- C: for growth – blue chip stocks, real estate.
- D: to get rich – risky investments, IPOs.

In financial literature, it is common to find an asset allocation strategy such as the example given in the figure above. People often consider their investments in separate "mental accounts," with each account representing a portfolio for different purposes. From the point of view of the combined assets, this view causes investors to misperceive risk. For example, emerging market equity funds or hedge fund investments are usually considered as "high risk", but they may reduce risk on a portfolio basis if their correlations with the other assets are low. Secondly, people may "diversify" within a narrow context, e.g. within our MPF contributions, whereas this is not a true diversification. Whether this kind of strategy is advisable depends on our benchmark for what is "optimal" (see section A1.3.1. below).

A1.2.5. Imperfections in the markets

Actions of noise traders

A general assumption in many finance theories is that people would always act rationally. But evidence from the markets show that there are bound to be *noise traders* – many investors who may not be rational, but prices would be affected by their actions. If one wants to be successful in trading, one has to be able to assess how these people act.

The following game was suggested by Richard Thaler (Nobel Prize winner and an economics professor from the University of Chicago), and the competition was held in April 1997 by the Financial Times newspaper, with a top prize of two first class return air tickets between London and New York/Chicago. Those entering the competition are asked to choose a whole number between 0 and 100. The winning entry would be the number which is closest to 2/3 of the average entry.

In order to win the game, one needs to understand how other people behave. Suppose you think the average of all the entries is 20. You should pick 14 as the winning number (approx. =20x2/3). But if everyone has the same thinking, the winning number should be 10 (=14x2/3), and so on. If this process is repeated, the logical winning entry should be 1 (or 0, depending on the rounding). With a group of well-educated readers of the Financial Times, the winning number was 13, which means most people made mistakes, or they anticipated that some people would not follow the logical reasoning.

The limits of arbitrage

Mature markets are supposed to be efficient, and any arbitrage opportunity would be eliminated quickly (refer to section A1.1.1 above). A penetrating analysis was developed in a paper by Shleifer & Vishny in 1997. Consistent with the arbitrage pricing theory, all that is needed to prevent arbitrage is for the existence of a relatively small number of highly specialized professional investors (or arbitrageurs) (also known as *smart money*). On the other hand, there are lots of "noise traders" in the market who are ignorant about the markets. In times of financial crisis, when prices diverge significantly from fundamental values, the arbitrageurs may not be able to borrow enough additional funds to carry out the strategy. Because of trading limits, arbitrageurs are often forced to close out early because of lack of funds, especially when the portfolio suffers a short term mark-to-market loss before the arbitrage profit can be realized (also refer to section 7.1.2) Therefore arbitrage would not be able to bring markets back to equilibrium within a short period of time – a necessary assumption in theories of efficient markets. While the paper attracted much academic interest, the authors became famous when the financial crisis occurred in 1998 and events unfolded exactly as predicted. The high profile hedge fund, Long Term Capital Management, was brought down primarily by the reason given above.

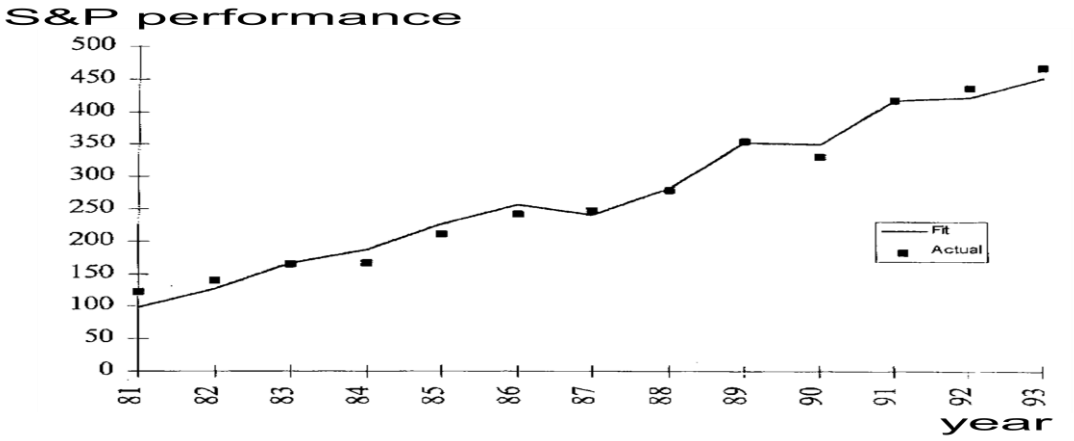
Market inefficiency

Any inefficiencies in the market are supposed to be short-lived, and they should soon be brought back into equilibrium. However, because of the judgment biases and psychological traits, market could be inefficient in the long run, meaning that: (i) abnormal trends could be observed; and (ii) asset prices may deviate from their fundamental values. Since investment decisions are consistently subject to these biases, deviations from fundamental values could remain in place for long periods of time. As a result, if we pursue an arbitrage strategy hoping that long term equilibrium may be established, it is possible that trends may not revert back to its mean (see sections 9.2.2. and 9.2.3.)

A1.2.6. Response from traditional finance

Given the evidence shown above, the Efficient Market Hypothesis would still have many supporters. Many arguments were proposed to counter the criticisms from behavioral finance. For example, Fama (1998) claims that market efficiency survives the challenge from behavioral finance theories. There may be as many occasions of people acting contrary to what behavioral finance theories predict would happen. Anomalies do not seem to repeat and they could be due to random effects. Furthermore, no consistently profitable strategy could be devised, and at the time of writing there is no testable alternative hypothesis based on behavioral theories (but see section A1.3.2. and A1.3.3.) Finally, Fama argues that while markets could be inefficient for short periods, they would "self adjust."

The existence of anomalies indicates a supposedly abnormal profitable trading strategy, but it is very possible that people have wrongly identified a backward looking strategy as successful. In particular, data mining is a potential problem.



A famous informal example was given by Leinweber (1997). The plot above shows the performance of the S&P 500 index from 1981 to 1993. Using a data set from the United Nations, he found the following three factors:

i. Bangladesh butter production;
ii. cheese production of Bangladesh + US;
iii. world sheep population.

Together they explained 99% of the movement of the annual closing prices of S&P between 1981-1993. (If only one factor is used, Bangladesh butter production produces the best fit.) Of course, it doesn't make sense, and there is no reason why these factors can be used to predict the movements of the S&P 500 in years after 1993.

The above shows an example of the confusion between correlation (a statistical concept) and causality (explanatory power). In the financial world in the United States, two cheeky indicators have been suggested to predict the performance of the stock market. With the hemline indicator ("bull markets and bare knees" theory), stock market returns are supposed to be good when miniskirts are in fashion. Even some sociological reasons are given to explain the validity of this claim. Another indicator comes from the prominent American football event, the Super Bowl. A victory by an NFL team predicts a bull market, whereas a victory by a former AFL team is bad. These indicators seem to provide a reasonable fit to the historical data, but highly correlated movements could also be a result of *data mining.* Finally, an unobserved external factor could also be a cause of seeming anomaly.

Stock market bubbles are recurrent events in the history of financial markets, and they have been used as counter-arguments to market efficiency. Basically, if markets are efficient, information should be dissimulated gradually and there should not be sudden changes of views of investors, which would lead to quick drops. A famous case was October 1987, when the US stock market lost about 1/3 of its value from early to mid-October with essentially no change in the general economic environment. Many people questioned how the market could be efficient both at the start and end of October. However, it was noted that the US interest rate had increased sharply two months prior to October 1987, which could lead to big changes in valuation, but many people had been unaware of the changes. In the next major crisis (Y2K and internet bubble), even at the height of the crisis in 2000, no obvious arbitrage opportunity was available. It was suggested that the market could be "temporarily inefficient" as too much new capital flowed to internet related stocks, but equilibrium was resumed after a brief period.

A1.2.7. Lessons learned

Given that we have evidence of all the fallbacks mentioned above, a valid question to ask is: if we can become a rational investor (e.g. use a computer to help us), would it be possible that we take advantage of all the irrational investors outside and earn more money? Unfortunately the answer is usually NO. As a smart investor, one could: (i) distinguish luck from skill; (ii) recognize the mistakes of other traders; (iii) know that only some risks are worth taking. We could avoid the problems and make less mistakes (and should lose less money), but there is no simple way to make money out of it. Despite all these human fallacies, the general consensus is that markets are more or less efficient!

To recap what we have learned so far, if the markets are (mostly) efficient, we should invest in index funds because:

- they are low cost;
- they should beat most fund managers and stock pickers in the long run; and
- active allocation would not achieve anything.

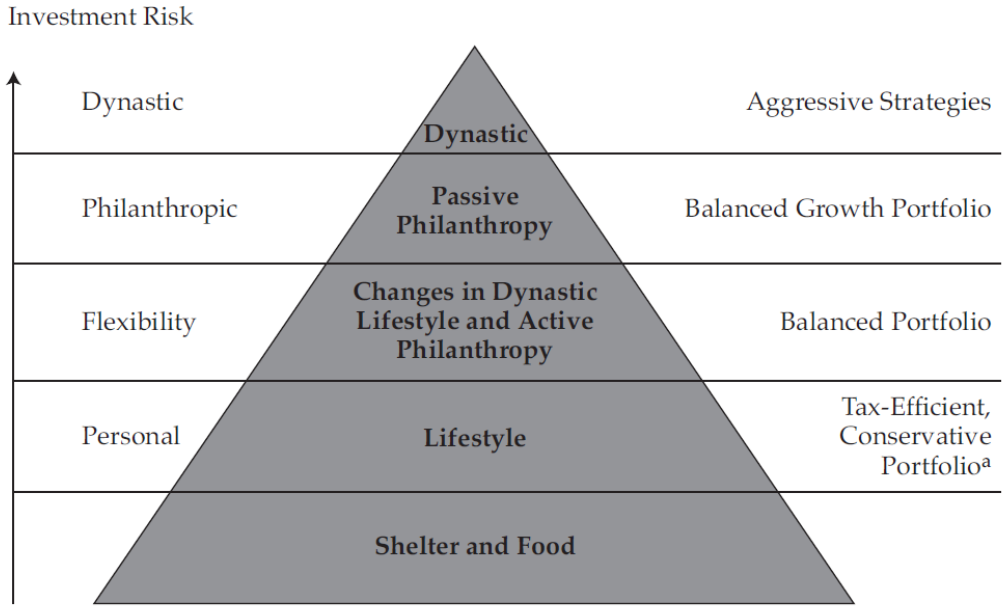
However, it would be impossible to "get rich quick" because one could only get an "average" performance every year. Furthermore, investing like this would be seen as boring, because in gatherings there are bound to be friends telling stories who boast that some of their stocks doubled or tripled. On the other hand, the long run performance is almost always superior. As a final caution, stock markets are often over-rated (c.f. Shiller (2000)) and judgment and experience can still turn out to be useful.

A1.3. Recent models with behavioral finance considerations

A1.3.1. Goal-based wealth management

MPT and efficient portfolios assume that investors would manage all their assets as a single portfolio, taking into account of the covariance between the asset returns. Any decision on diversification should be made by considering the portfolio as a whole, and the "mental account" picture given in section A1.2.4 above is not supposed to be optimal. More recently, there has been a move towards "goal-based wealth management," where insights from behavioral finance are combined with portfolio theory. Instead of a top-down approach of identifying an optimal portfolio based on risk-return characteristics alone, this approach would try to identify the various goals of an investor, and acknowledges the fact that different risk aversion levels would be appropriate for different goals, while the resulting asset allocation should not be based on a single and comprehensive portfolio. While each part of the portfolio can still be optimized using models such as the MPT, the overall goal is a big deviation from the MPT. Basically the "mental account" picture is accepted as being more appropriate in formulating portfolios, as this is seen as a more valid response to what the investor wants.

Exhibit 1. The Behavioral Finance Portfolio Pyramid



[a]Risk taken only to preserve long-term purchasing power.

Source: Meir Statman, "What Do Investors Want?" *Journal of Portfolio Management*, vol. 30, no. 5 (30th anniversary 2004):153–161.

Taken from: Jean Brunel, "Goal-based Wealth Management in Practice" (2012).

Instead of focusing on a single statistical parameter (variance or covariance) as the measure of risk, many alternative models have been developed based on decisions that concentrate on "ruins" or bad outcomes (sometimes known as "safety first" models). Examples of models of this type include:

- Roy(1952) – Minimize $\text{Prob}(W<s)$, where Prob() refers to a probability distribution, W is the final wealth level, and s is a subsistence level, i.e. a minimum acceptable wealth level.
- Kataoka(1963) – Maximize s, subject to $\text{Prob}(W<s)<=\alpha$ (a typical value of α is 5%)
- Telser(1955) – Maximize $\text{E}(W)$, subject to $\text{Prob}(W<s) =\alpha$, where E() represents the expected value.

A formal framework known as SP/A was introduced by Lopes (1987) as a model for decision making. The investor's choice is driven by three factors: security (S), potential (P), and aspiration (A). Aspiration refers to the specific target final wealth level as defined by the investor. Furthermore, there are two emotional drivers: investors are driven by fear to secure their wealth, and they are also willing to take risk to increase potential gains. In order to incorporate these factors, the choices are expressed through a transformation of the probability distribution. Probabilities associated with the worst outcomes and the best outcomes are both adjusted, specific to the goals of each investor. The adjusted probability distribution will significantly affect the results of any optimization process.

A1.3.2. Behavioral portfolio theory (BPT)

Formulated by Shefrin and Statman (2000), BPT makes use of the SP/A decision framework and prospect theory of Kahneman and Tversky (1979), as well as incorporating features of mental accounts. The BPT optimal portfolio satisfies:

$$\max E_h(W) \text{ subject to } P(W < A) < \alpha$$

where W is the final wealth distribution of the investor, A is the aspiration level and α is the acceptable probability of ruin. With mental accounts, the multiple-accounts version called BPT-MA allows for different aspirations and risk tolerance in more than one portfolio for the same investor, and the final portfolio mix is the sum of the utility of each portfolio.

One of the key features of the theory is the definition of the expected wealth distribution. It can be represented by a set of probabilities, but investors often make use of decision weights, and overweight probabilities of extreme events. In the Cumulative Prospect Theory, investors determine the subjective value of each outcome via a value function, where utility is derived from changes relative to a reference point. Based on experimental evidence, Tversky and Kahneman (1992) proposed the following functional form for the value function:

$$v(x) = \begin{Bmatrix} x^{\alpha} & \text{if } x > 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0 \end{Bmatrix}$$

$$\alpha = \beta = 0.88, \lambda = 2.25$$

This value function is concave over gains and convex over losses, and the parameter λ determines the degree of loss aversion (c.f. the figure in section A1.2.3.)

Results from BPT are often compared with MPT, but different conclusions are drawn by different authors. Shefrin and Statman (2000) show that the BPT optimal portfolio is often different from the Markowitz optimal portfolio. Das et al. (2004) have suggested a mental accounting (MA) framework which combines BPT with MPT, and show that the MA optimal portfolio always lies on the MPT efficient frontier. This is in sharp contrast with the results obtained by Alexander and Baptista (2011). Pfiffelmann et al. (2015) use US stock prices from 1995-2011 to compare the BPT optimal portfolio with MPT's results, and finds that in more than 70% of cases the BPT optimal portfolio lies on the MPT efficient frontier.

A1.3.3. Popularity asset pricing model

CAPM expresses investors' preferences solely in terms of risk, which is captured by the parameter beta. However, it is suggested that other asset characteristics such as liquidity and tax issues, environment, social and governance (ESG) factors, as well as behavioral (irrational) factors and so forth, may also impact the security returns.

Ibbotson et al. (2018) propose that a new model is needed to provide a better explanation to the existence of premiums due to these and other factors. Known as Popularity Asset Pricing Model (PAPM), CAPM is extended to include security characteristics that different investors regard differently. The expected return on each security is a linear function of its beta and its popularity loadings which measure the popularity of the security based on its characteristics relative to those of the beta-adjusted market portfolio. Each investor holds a different customized portfolio of risky assets based on his/her attitude towards security characteristics.

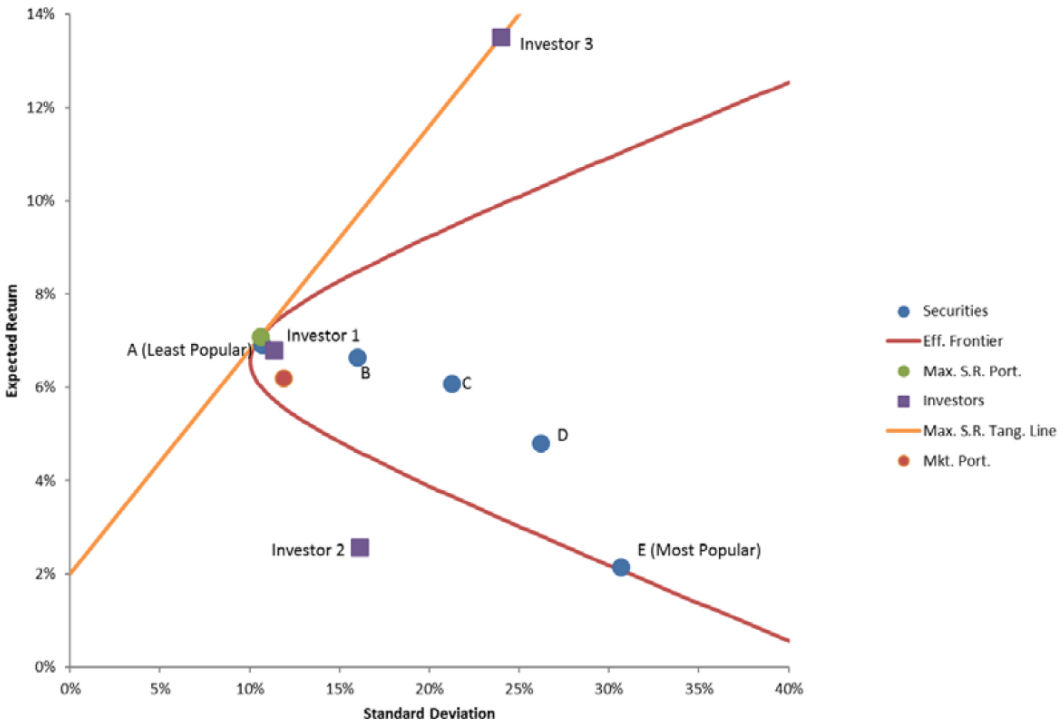


Figure taken from Kaplan (2017).

A typical efficient frontier is shown in the above diagram. Compared to the MPT, a key difference is that the market portfolio no longer lies on the efficient frontier. However, a major difficulty of the model is that all investors need to agree on what the characteristics of the securities are (i.e. the popularity factors), and it is not obvious how such factors can be determined.

Additional references

Introductory

Bernstein, W.J. (2002). *The Four Pillars of Investing*. McGraw Hill.

Kahneman, D. (2011). *Thinking, Fast and Slow*. Penguin Books.

Lo, A.W. (2017). *Adaptive Markets: Financial Evolution at the Speed of Thought*. Princeton University Press.

Malkiel, B. (2019). *A Random Walk down Wall Street*, 12th edition. Norton.

Nofsinger, J.R. (2002). *The Psychology of Investing*. Prentice Hall.

Shiller, R. (2000). *Irrational Exuberance*. Princeton University Press.

Thaler, R. (2015). *Misbehaving.* Penguin Books.

Academic References

de Bondt, W. & Thaler, R. "Does the Stock Market Over-react?" *Journal of Finance* 40:793-805, 1985.

Brunel, J. "Goal-based Wealth Management in Practice." *CFA Institute Quarterly Conference Proceedings*, Vol 29, no. 1, March 2012.

Chhabra, A. "Beyond Markowitz: A Comprehensive Wealth Allocation Framework for Individual Investors." *Journal of Wealth Management*, 7(4):8-34, Spring 2005.

Das, S.R, Markowitz H., Scheid J. and Statman M. "Portfolio Optimization with Mental Accounts." *Journal of Financial and Quantitative Analysis*, 45(2):311-334, April 2010.

Fama, E. "Market Efficiency, Long-term Returns, and Behaviour Finance." *Journal of Financial Economics* 49 (3):283-306, 1998.

Ibbotson, R.G., Idzorek T.M., Kaplan P.D. and Xiong J.X. *Popularity: A Bridge between Classical and Behavioral Finance*. CFA Research Foundation, 2018.

Kahneman D. & Tversky, A. "Prospect Theory: an Analysis of Decision under Risk." *Econometrica* 47:263-291, 1979.

Kaplan, P.D. "A Popularity Asset Pricing Model." Working paper, 2017.

Pfiffelmann, M., Roger T. and Bourachnikova O. "When Behavioral Portfolio Theory Meets Markowitz Theory." *Economic Modelling*, October 2015

Shefrin, H. *Beyond Greed and Fear.* Harvard Business School Press, 2000.

Shefrin, H. and Statman, M. "Behavioral Portfolio Theory." *Journal of Financial and Quantitative Analysis*, 35:127-151, 2000.

Shleifer, A. & Vishny, R. "The Limits of Arbitrage." *Journal of Finance* 52:35-56, 1997.

Tversky, A. & Kahneman D. "Judgment under Uncertainty: Heuristics and Biases." *Science* (185):1124-1131, 1974.

Tversky, A. & Kahneman D. "Rational Choice and the Framing of Decisions." *Journal of Business* 59, no. 4, part 2:251-278, 1986.

Appendix: Questionnaire on behavioral finance

Instruction:
In each question, unless otherwise stated, choose either (A) or (B). This is not a mathematical exercise and there is no right or wrong answer for each question. Pick your answers based on "gut feeling", i.e. what you would actually choose when you are faced with a real decision. The whole set should be completed in less than 5 minutes.

1. A) A sure gain of $3000
   B) 80% chance of winning $4000, 20% chance of winning 0

2. Of all the members of the United Nations, how many are African countries?
   Is it higher or lower than 55%? Higher / Lower
   What is your estimate?

3. A) 25% chance of winning $3000, 75% of winning 0
   B) 20% chance of winning $4000, 80% of winning 0

4. 5 days before the match, you bought a ticket to watch Brazil vs HK football team for $800 (you don't want to pay $1200 or $1500 to buy the expensive ticket because you may think that it is not worth the price). When you reached the stadium on the day of the match, you found that you have lost the ticket. Would you buy another $800 ticket at the gate and go to see the match?

   Yes / No

5. A) A sure loss of $7500
   B) 75% chance of losing $10000, 25% chance of losing 0

6. You own two stocks and each is worth $10000 now. Stock A was bought at $8000 and stock B was bought at $11000. You need money and have to sell one of them to get $10000. Which one would you likely sell?

   Stock A / Stock B / Don't know

7. A) 25% chance of winning $2400, 75% chance of losing $7600
   B) 25% chance of winning $2500, 75% chance of losing $7500

8. A) A sure gain of $2400
   B) 25% chance of gaining $10000, 75% chance of gaining 0

9. A) 80% chance of losing $4000, 20% chance of losing 0
   B) A sure loss of $3000

10. You want to see a concert and tickets at $800 are available at the concert hall. When you are ready to pay for the ticket, you discovered that you have lost the big "lai see 紅包" of $800 that you received earlier. Would you still buy a ticket and see the concert?

Yes / No

- END -